

THE DIGITAL IDENTITY CARD

How Websites & Apps Remember Who You Are!

Planrs Academy Master Course • Computer Science & Cyber Safety for Kids

Think About It...

Have you ever wondered why Amazon, Netflix, or YouTube doesn't ask for your password every single time you click on a new video or page?

"Imagine if every time you turned a page in a textbook, the book locked itself and demanded: 'Who are you? Prove you're allowed to read this page!' That would be super annoying, right?"

Websites use a secret mechanism called **Sessions** and **Tokens** to remember you after you log in once!

1. IDENTIFICATION VS. AUTHENTICATION

Identification

What it is: Simply telling someone who you are.

Real-World Example: Saying "I am a student at Planrs Academy" when the teacher calls your name.

Digital Example: Typing your Username or Email address.

Authentication

What it is: Proving that you are actually who you say you are.

Real-World Example: Showing your School ID card with your photo and Principal's official stamp.

Digital Example: Entering your Secret Password or Scanning your Fingerprint.

2. HOW WEBSITES REMEMBER YOU: REAL-WORLD ANALOGIES

Sessions: The Amusement Park Wristband

Imagine you visit an amusement park like Wonderla or Disneyland. At the entry gate, you pay for your ticket and prove your identity once. Instead of asking for your paper ticket at every single roller coaster, the gatekeeper gives you a **Wristband**.

- **Pay once:** Show your ticket at the front gate (Login).
- **Get a Wristband:** Wear it on your wrist (Session Created).
- **Ride everything:** Every ride operator sees your wristband and lets you right in without re-checking your ticket!

Tokens & JWT: The Metro Express Pass

Think about the Metro Token or Smart Card in cities like Delhi or Bengaluru. That small plastic token holds specific digital information: *Where you entered, where you're allowed to travel, and when it expires.*

In web development, we use **JSON Web Tokens (JWT)**. These are lightweight digital passes containing encoded details (called **Claims**) that tell the server exactly what you can access!

Real-World National ID: The Aadhaar System

In India, citizens use a secure digital identification framework called **Aadhaar**—a unique 12-digit identification system managed by UIDAI. It serves over 1.3 billion people, allowing secure identity verification both online (via OTP / Biometrics) and offline.

INTERACTIVE WORKSHEET & STEP-BY-STEP FLOW

Understanding how data flows between your browser and the website

3. THE 3-STEP DIGITAL KEY FLOW

1. **Step 1: Login (Authentication)** — You enter your Username and Password on the login page. Your browser sends this securely to the web server.
2. **Step 2: Key Issuance (Token Generation)** — The server verifies your credentials. If correct, it generates a special encrypted **Digital Key (JWT)** and sends it back to your browser.
3. **Step 3: Automatic Access (Session Request)** — Whenever you click a new page, watch a video, or add an item to your shopping cart, your browser automatically attaches this Digital Key in the background. No extra typing needed!

4. SAFETY & CYBER SECURITY RULES

✓ The Good Side (Convenience)

Saves time! You don't have to re-type passwords on every click. Apps like Hotstar or Spotify can resume your movie or music right where you stopped.

⚠ The Security Rule (Safety)

Your session token is like a physical key. If you use a public computer (like at school or a library) and walk away without clicking **LOG OUT**, anyone else can use your active key!



Activity: Design Your Own Custom Digital JWT Token!

Imagine you are a lead web designer at Planrs Academy. Create your custom access pass below!

[Draw your Custom Token / Badge / Shield here]

Token Specs:

1. Shape: _____
2. User Name: _____
3. Age Claim: _____
4. Access Level:
[] Member [] VIP Admin
5. Expiration Time: _____

Fun Facts to Remember!

- **Smart Cards:** Metro tokens are gradually being replaced by contactless RFID Smart Cards—which act as long-term reusable tokens!
- **Auto-Save:** Without Sessions, every video you watch on JioCinema or Hotstar would restart from 0:00 if you refreshed the browser!

STUDENT PRACTICE & ASSESSMENT QUESTIONS

Test your knowledge on Digital Identities, Sessions, Tokens, and Cyber Safety

Instructions: Answer all questions below in your notebook or provided blank lines.

PART A: MULTIPLE CHOICE QUESTIONS

1. What is the main difference between "Identification" and "Authentication"?

- A) Identification is proving who you are; Authentication is stating your name.
- B) Identification is stating who you are; Authentication is proving it with credentials like a password.
- C) Both words mean the exact same thing in computer science.
- D) Authentication only happens at amusement parks.

2. What real-world item best represents how a web "Session" works when visiting multiple pages?

- A) A library card left at home.
- B) An amusement park wristband given after showing your entry ticket.
- C) A paper receipt you throw in the trash immediately.
- D) A school textbook.

3. What does JWT stand for in digital token security?

- A) Java Web Tool
- B) Joint World Ticket
- C) JSON Web Token
- D) Just Website Text

PART B: SHORT ANSWER & SCENARIO QUESTIONS

4. Explain why it is dangerous to leave your account logged in on a public computer at a library or school lab. What should you always do before leaving?

5. Describe how streaming platforms like Hotstar or Netflix use "Sessions" and "Tokens" to remember the exact minute of a movie you were watching yesterday.

6. Match each item on the left with its correct analogy or description on the right:

Terms:

1. Web Session
2. Digital Token (JWT)
3. Aadhaar Card
4. Password

Analogies / Descriptions:

- ☐ Metro Pass with destination info
- ☐ Proof used for Authentication
- ☐ Amusement Park Wristband
- ☐ Unique 12-Digit Indian ID

7. Complete the 3-step sequence of getting and using a digital key:

Step 1: User enters credentials to _____.

Step 2: Web Server verifies and issues a _____.

Step 3: Browser presents key on each request for automatic _____.